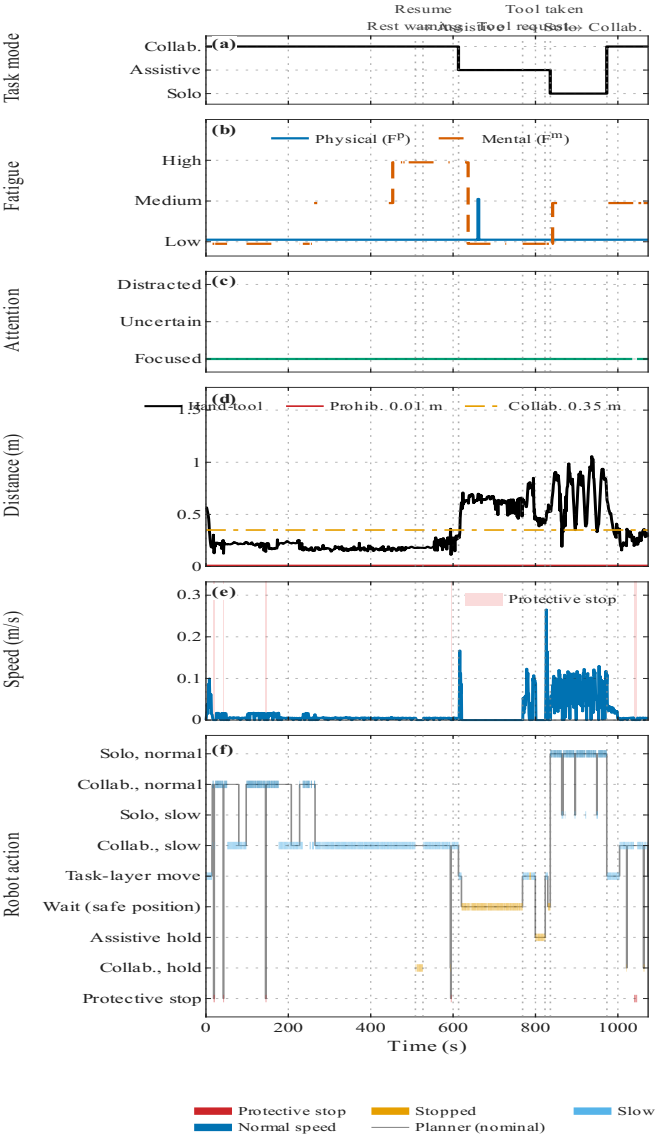

Resume
Rest warning
Tool taken
Assistive → Tool request
Solo → Collab.
Task mode
(a)
Collab.
Assistive
Solo
Fatigue
(b)
Physical (F^P)
Mental (F^m)
High
Medium
Low
Attention
(c)
Distracted
Uncertain
Focused
Distance (m)
(d)
Hand-tool
Prohib. 0.01 m
Collab. 0.35 m
1.5
1
0.5
0
Speed (m/s)
(e)
Protective stop
0.3
0.2
0.1
0
Robot action
(f)
Solo, normal
Collab., normal
Solo, slow
Collab., slow
Task-layer move
Wait (safe position)
Assistive hold
Collab., hold
Protective stop
0
200
400
600
800
1000
Time (s)
Protective stop
Normal speed
Stopped
Planner (nominal)
Slow